

The cotype–cotype conjecture under the approximation property

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Abstract

We prove the cotype–cotype conjecture under the ordinary approximation property. A nonzero real Banach space with this property is K -convex if and only if both the space and its dual have finite Rademacher cotype, possibly with different exponents.

1 The characterization

The cotype–cotype problem asks whether finite Rademacher cotype of a Banach space and its dual forces K -convexity. Without an approximation hypothesis the assertion is false. We prove the characterization under the ordinary approximation property.

All spaces in this paper are real, and X^* denotes the continuous dual of X . Recall that X has the *approximation property* (AP) if, for every compact set $M \subset X$ and every $\delta > 0$, there is a bounded finite-rank linear map $S : X \rightarrow X$ such that

$$\sup_{x \in M} \|Sx - x\| < \delta.$$

The norms of these maps need not have a common bound. The *bounded approximation property* (BAP) requires, in addition, a constant $\lambda < \infty$ such that all the maps needed for these approximations can be chosen with $\|S\| \leq \lambda$.

For $2 \leq q < \infty$, let $C_q(Y)$ be the least constant such that

$$\left(\sum_{i=1}^s \|y_i\|^q \right)^{1/q} \leq C_q(Y) \left(\mathbb{E}_\varepsilon \left\| \sum_{i=1}^s \varepsilon_i y_i \right\|^2 \right)^{1/2}$$

for every finite family in Y , where the ε_i are independent uniform signs. The space Y has *finite Rademacher cotype* if $C_q(Y) < \infty$ for some finite q .

The *Rademacher projection* on X -valued functions on a sign cube retains their first-order coefficients:

$$(\mathcal{R}_s f)(\varepsilon) = \sum_{i=1}^s \varepsilon_i \mathbb{E}_\eta [\eta_i f(\eta)].$$

Write

$$K(X) = \sup_{s \geq 1} \|\mathcal{R}_s : L_2(\{-1, 1\}^s; X) \rightarrow L_2(\{-1, 1\}^s; X)\|.$$

Here L_2 uses uniform probability measure. By definition, X is K -convex when $K(X) < \infty$.

Theorem 1.1. *Let X be a nonzero real Banach space with the ordinary approximation property. Then X is K -convex if and only if*

$$C_q(X) < \infty \quad \text{and} \quad C_r(X^*) < \infty \quad \text{for some } q, r \in [2, \infty).$$

The exponents q and r may differ. The forward implication holds for every real Banach space, without an approximation hypothesis.

A space X has *Rademacher type p* , $1 < p \leq 2$, if for some $D < \infty$,

$$\left(\mathbb{E}_\varepsilon \left\| \sum_i \varepsilon_i x_i \right\|^2 \right)^{1/2} \leq D \left(\sum_i \|x_i\|^p \right)^{1/p}$$

for all finite families. We say that X contains the ℓ_1^s 's uniformly if the isomorphism distortions of such copies inside X are bounded independently of s .

The classical theorems of Maurey–Pisier and Pisier [7, 12] identify K -convexity with nontrivial type:

$$\begin{aligned} K(X) < \infty &\iff X \text{ has Rademacher type } p \text{ for some } p > 1 \\ &\iff X \text{ does not contain the } \ell_1^s \text{'s uniformly.} \end{aligned} \tag{1.1}$$

These equivalences require no approximation hypothesis; precise formulations are [15, Theorem 4.40(i)] and [10, Theorem 1.5]. The finite-cube definition above agrees with the infinite-product definition in [10, Definition 1.1], by density of functions depending on finitely many coordinates.

The characterization is uniform when the cotype data are fixed. Its finite-dimensional form also supplies a class of convex bodies for which ordinary metric-entropy duality has dimension-free constants. For an origin-symmetric convex body $B \subset \mathbb{R}^n$, write X_B for $\mathbb{R}^n$ with unit ball B , and B° for its polar. Here a convex body is compact with nonempty interior, and $N(S, T)$ is the least number of translates of T needed to cover S .

Corollary 1.2 (Uniform cotype bounds and entropy duality). *Fix $q, r \in [2, \infty)$ and $C, D \in [1, \infty)$. There is a finite constant $\kappa = \kappa(q, r, C, D)$ such that every nonzero real Banach space X with AP and*

$$C_q(X) \leq C, \quad C_r(X^*) \leq D$$

satisfies $K(X) \leq \kappa$. Consequently there are $a, b \geq 1$, depending only on q, r, C, D , such that for every $n \geq 1$, all origin-symmetric convex bodies $K, L \subset \mathbb{R}^n$, and every $t > 0$,

$$\frac{1}{b} \log N(L^\circ, atK^\circ) \leq \log N(K, tL) \leq b \log N(L^\circ, a^{-1}tK^\circ), \tag{1.2}$$

provided either X_K or X_L satisfies the displayed primal and dual cotype bounds.

The constants are existential; no bound on the AP approximants is required. The entropy conclusion uses the bounded- K duality theorem of Artstein–Milman–Szarek–Tomczak–Jaegermann [1, Theorem 5], in the covering-number form of [8, Theorem 1.1 and Section 2]. We prove the corollary at the end of Section 5.

1.1 History and scope

Type and cotype express the geometry of a Banach space through the size of random signed sums. Their relation to uniform finite-dimensional subspaces was developed by Maurey and Pisier [7]. Elementary duality passes a type estimate to a cotype estimate on the dual, but recovering type from cotype requires further information. Maurey and Pisier introduced K -convexity as the setting for this duality question; Pisier's holomorphic-semigroup theorem subsequently identified it with nontrivial type [7, 10, 12]. The question here asks whether finite cotype on both sides, together with approximation, already puts the space in that setting.

Approximation assumptions have played a substantial role in these questions. Enflo's counterexample showed that a separable reflexive Banach space can fail AP [3]. A Schauder basis gives uniformly bounded finite-rank approximations, but the converse implications in the hierarchy

$$\text{Schauder basis} \implies \text{BAP} \implies \text{AP}$$

fail: Figiel and Johnson separated AP from BAP [4, p. 199], and Szarek constructed a superreflexive space with BAP but no Schauder basis [17, Theorem 1.1]. Thus an argument using ordinary AP must accommodate approximants whose norms have no common bound.

For cotype two on both sides, Pisier established the stronger conclusion that an AP space is isomorphic to a Hilbert space [9, Corollaries 2.9 and 2.16]. His subsequent work proved K -convexity under a restriction on the cotype exponents, first with a basis or a dense increasing family of uniformly complemented finite-dimensional subspaces, and then with BAP [11, Theorem 1, Remarks 2 and 9]. He asked whether the exponent restriction could be removed [11, Remark 2(iii)]. Szarek and Tomczak-Jaegermann discuss this question as the *cotype-cotype conjecture* in connection with uniformly complemented finite-dimensional subspaces [18]. Gupta, Misra, and Ray state the BAP formulation in [6, Conjecture 1.5]. Theorem 1.1 proves the characterization for all finite cotype exponents under ordinary AP.

The assertion with no approximation hypothesis is false. Pisier's construction embeds every cotype-two space into a space whose primal and dual both have cotype two [13, Theorem 3.2]. Starting with ℓ_1 therefore gives a space which is not K -convex; these examples fail AP [13, Remark 3.6].

Another application concerns a uniform tensor estimate. Given a sequence $\mathcal{E} = (E_n)$ with $\dim E_n = n$, one asks whether every map $A : E_m \rightarrow E_n^*$, tensored with the identity of F , is bounded from the injective tensor norm to the projective tensor norm by one constant times $\|A\|$, independently of m, n, A . This is the Grothendieck-pair condition, stated formally in Section 6. Gupta–Misra–Ray prove that it forces finite cotype of F and F^* , and record Pisier's conjecture that for infinite-dimensional BAP spaces F the estimate then also holds with ℓ_2 in place of F [6, Definition 1.2, Theorem 1.3 and Conjecture 1.4]. Their deduction through complemented Euclidean subspaces [6, Corollary 4.8], combined with Theorem 1.1, gives this conclusion under ordinary AP. This uniform operator estimate differs from Grothendieck's original injective/projective tensor-product conjecture, disproved in [13], and from the GT-pairs of [16, Section 6].

Rademacher and Walsh expansions connect probabilistic moment estimates with Banach-space geometry. Kahane's inequality compares moments of vector-valued Rademacher sums [15, Theorem 8.1]; Borell extended moment comparison to Banach-valued Walsh polynomials of bounded degree [2]. The moment lifts below combine the Rademacher comparison with quotient duality and Hahn–Banach. The scalar coefficient lemma uses an additional feature of Fourier indices: their rank over $\mathbb{F}_2$, which determines the number of independent Walsh factors.

1.2 The mechanism of the proof

The proof has two parts. First, we establish a uniform decay estimate for the normalized Walsh transforms of finite-rank operators $T : G \rightarrow F$: apply T to a vector family and then multiply by the matrix $(2^{-n}(-1)^{u \cdot v})_{u,v \in \mathbb{F}_2^n}$, with the dot product computed modulo two and probability-normalized L_2 norms. Cotype estimates for G^* and F at one sufficiently large, fixed tuple length suffice. The decay estimate, relative to $\|T\|$, is independent of the operator rank and of the spaces; Theorem 4.3 states the precise result.

A fixed finite-rank operator already has a Walsh bound obtained by factoring through a Hilbert space, but its constant depends on the operator. To remove that dependence, we show that a slowly decreasing Walsh norm forces another substantial norm at nearly twice the dimension. A moment lift represents a tuple (x_i) by a bounded sign-indexed family (P_ε) with $\mathbb{E}_\varepsilon \varepsilon_i P_\varepsilon = ax_i$, where $a > 0$ is controlled by cotype. Such lifts and finite-group averaging turn the slow decrease into bounded scalar functions with prescribed row and column moments. Their Fourier expansion is indexed by binary matrices and must contain an index of rank at least two over $\mathbb{F}_2$. Such an index produces two independent Walsh factors. Iteration would then contradict the Hilbert-space bound for the same fixed operator. This proves uniform decay.

Second, we transfer that decay to the identity on X using only AP. For each finite-dimensional $E \subset X$, we construct a compact map $J : G \rightarrow X$ of uniformly bounded norm and a contraction $\iota : E \rightarrow G$ with $J\iota x = x$. The dual of G has a controlled moment estimate (Lemma 5.1). The construction uses moment lifts in X , rather than assuming that the quotient E^* inherits the cotype of X^* . AP gives finite-rank composites $S_j J$ converging to J in operator norm. Uniform Walsh decay passes to J , and then to every finite Walsh input in X . Uniform copies of ℓ_1^s would contradict that decay.

Section 2 introduces Walsh norms and the moment lifts. Section 3 proves the scalar coefficient lemma, and Section 4 turns it into uniform operator decay. Section 5 constructs the compact domains and proves Theorem 1.1 and Corollary 1.2. Section 6 gives the tensor-norm consequence.

2 Walsh transforms and moment lifts

We first introduce the operator norms that measure Walsh decay, then derive the moment lifts needed to compare those norms at different dimensions. Throughout, Banach spaces are real. All unspecified averages on finite sets are uniform, and distinct averaging variables are independent. The notation B_Y means the closed unit ball of Y . For a finite probability space Ω , the space $L_p(\Omega; Y)$ has its probability-normalized norm. By contrast, $\ell_2^m(Y)$ has norm $(\sum_{i=1}^m \|y_i\|^2)^{1/2}$. Duality on finite probability spaces uses the expectation pairing; in particular,

$$\begin{aligned} L_2(\Omega; Y)^* &= L_2(\Omega; Y^*), \\ L_1(\Omega; Y)^* &= L_\infty(\Omega; Y^*), \quad L_\infty(\Omega; Y)^* = L_1(\Omega; Y^*). \end{aligned}$$

2.1 Walsh norms

Let $V_n = \mathbb{F}_2^n$, with its standard dot pairing, and put $\chi(u, v) = (-1)^{u \cdot v}$. For $T : G \rightarrow F$, define

$$(W_n^T x)(v) = \mathbb{E}_{u \in V_n} \chi(u, v) T x(u), \quad w_n(T) = \|W_n^T : L_2(V_n; G) \rightarrow L_2(V_n; F)\|.$$

We allow $n = 0$, so $w_0(T) = \|T\|$. Finite-sum duality gives the equivalent formula, with $x_u \in G$ and $y_v \in F^*$,

$$w_n(T) = \sup |\mathbb{E}_{u,v \in V_n} \chi(u, v) y_v(T x_u)|, \quad \mathbb{E}_u \|x_u\|^2 \leq 1, \quad \mathbb{E}_v \|y_v\|^2 \leq 1. \quad (2.1)$$

Lemma 2.1 (Basic Walsh estimates). *For every bounded $T : G \rightarrow F$:*

- (i) $2^{-n/2}\|T\| \leq w_n(T) \leq \|T\|$, and $w_n(T)$ is nonincreasing in n .
- (ii) If $T_\Omega : L_2(\Omega; G) \rightarrow L_2(\Omega; F)$ acts pointwise as T , where Ω is finite, then $w_n(T_\Omega) = w_n(T)$.
- (iii) If T has finite rank, there is $D_T < \infty$ such that

$$w_n(T) \leq D_T 2^{-n/2} \quad (n \geq 0). \quad (2.2)$$

Moreover, for $A, B : G \rightarrow F$,

$$|w_n(A) - w_n(B)| \leq \|A - B\|. \quad (2.3)$$

Proof. The upper bound follows from the triangle inequality and Cauchy–Schwarz; the lower bound follows by taking input supported at one u , scaled by $2^{n/2}$. On $V_{n+j} = V_n \oplus V_j$, first apply W_n^T in the first coordinate and then the normalized Walsh transform with the identity of F in the second coordinate. The latter has norm at most 1. Averaging the squared estimates in the unused coordinates proves monotonicity. Interchanging finite L_2 coordinates proves (ii), with constant functions on Ω giving the reverse inequality. For (iii), factor a finite-rank T as $G \rightarrow H \rightarrow F$, with H a finite-dimensional Hilbert space. Orthogonality of the Walsh matrix gives $\|W_n^{\text{Id}_H}\| = 2^{-n/2}$, proving (iii). Finally, $W_n^A - W_n^B = W_n^{A-B}$, so the upper bound in (i) and the reverse triangle inequality prove (2.3). $\square$

2.2 Fixed-length cotype estimates

Define $c_m(Y)$ to be the least constant in

$$\left(\sum_{i=1}^m \|y_i\|^2 \right)^{1/2} \leq c_m(Y) \left(\mathbb{E}_\varepsilon \left\| \sum_{i=1}^m \varepsilon_i y_i \right\|^2 \right)^{1/2}. \quad (2.4)$$

Thus, if Y has cotype q with constant $C_q(Y)$, then

$$c_m(Y) \leq C_q(Y) m^{1/2-1/q}. \quad (2.5)$$

Squaring (2.4) and averaging proves

$$c_m(L_2(\Omega; Y)) \leq c_m(Y). \quad (2.6)$$

The same observation applies to duals of these finite L_2 sums.

The following estimate is the L_1 -to- L_2 case of Kahane’s inequality; see [15, Theorem 8.1]. We include a stopping-time proof with an explicit, sufficient constant so that the subsequent moment lifts are self-contained.

Lemma 2.2 (Moment comparison). *For every finite Rademacher sum $S = \sum_{i=1}^m \varepsilon_i z_i$ in a Banach space,*

$$(\mathbb{E}\|S\|^2)^{1/2} \leq L \mathbb{E}\|S\|, \quad L = 24. \quad (2.7)$$

Proof. Put $M = \mathbb{E}\|S\|$ and $b = \max_i \|z_i\|$. Since $z_i = \mathbb{E}(\varepsilon_i S)$, we have $b \leq M$. The case $M = 0$ is immediate.

Let $S_j = \sum_{i \leq j} \varepsilon_i z_i$, and stop at the first j with $\|S_j\| > u$. Conditional on a stopped prefix z , the remaining sum is symmetric; at least one of $\|z + h\|$ and $\|z - h\|$ exceeds u . Hence

$$\mathbb{P}(\text{a stop occurs}) \leq 2\mathbb{P}(\|S\| > u).$$

For any fixed j , symmetry of the independent prefix similarly gives

$$\mathbb{P}\left(\left\|\sum_{i>j}\varepsilon_i z_i\right\|>v\right)\leq 2\mathbb{P}(\|S\|>v).$$

At the first stop, $\|S_j\|\leq u+b$. Decomposing according to its time and using independence of the remaining sum therefore yields

$$\mathbb{P}(\|S\|>u+b+v)\leq 4\mathbb{P}(\|S\|>u)\mathbb{P}(\|S\|>v).$$

Take $u=8M$. Markov's inequality and $b\leq M$ imply

$$\mathbb{P}(\|S\|>9M+v)\leq \tfrac{1}{2}\mathbb{P}(\|S\|>v).$$

Thus $\mathbb{P}(\|S\|>9jM)\leq 2^{-j}$ for $j\geq 0$, and integration over consecutive intervals of length $9M$ gives

$$\mathbb{E}\|S\|^2\leq 81M^2\sum_{j\geq 0}(2j+1)2^{-j}=486M^2<24^2M^2.$$

□

We will repeatedly use a simple consequence of quotient duality. If a bounded surjection $U:A\rightarrow B$ satisfies $\|U^*b^*\|\geq D^{-1}\|b^*\|$, then

$$\inf_{Ua=b}\|a\|\leq D\|b\|. \quad (2.8)$$

Indeed, $A/\ker U$ is isomorphic to B , and its quotient norm at b is

$$\sup_{\|U^*b^*\|\leq 1}|\langle b,b^*\rangle|.$$

In each application below surjectivity has an explicit bounded right inverse, so the quotient identification is immediate.

Lemma 2.3 (Bounded moment lifts). *Suppose*

$$c_m(G^*)\leq D_g, \quad c_m(F)\leq D_f, \quad D_g, D_f\geq 1.$$

Set

$$a=\frac{1}{2LD_g\sqrt{m}}, \quad b=\frac{1}{2LD_f\sqrt{m}}. \quad (2.9)$$

Then:

(i) For $x_1,\dots,x_m\in B_G$, there are $P_\varepsilon\in B_G$ such that

$$\mathbb{E}_\varepsilon\varepsilon_iP_\varepsilon=ax_i \quad (1\leq i\leq m).$$

(ii) For $y_1,\dots,y_m\in B_{F^*}$, there are $Z_\delta\in B_{F^*}$ such that

$$\mathbb{E}_\delta\delta_jZ_\delta=by_j \quad (1\leq j\leq m).$$

The same conclusions, with the same constants, hold after replacing G, F by finite L_2 sums of their copies.

Proof. The moment map

$$U : L_\infty(\{-1, 1\}^m; G) \longrightarrow \ell_2^m(G), \quad UP = (\mathbb{E}_\varepsilon \varepsilon_i P_\varepsilon)_{i=1}^m,$$

is onto: a right inverse is $(x_i) \mapsto \sum_i \varepsilon_i x_i$. Its adjoint sends (z_i^*) to $\sum_i \varepsilon_i z_i^*$. By Lemma 2.2 and (2.4),

$$\|U^*(z_i^*)\|_{L_1(G^*)} \geq (LD_g)^{-1} \left(\sum_i \|z_i^*\|^2 \right)^{1/2}.$$

Equation (2.8) bounds the infimal lift norm of (ax_i) by $LD_g a\sqrt{m} = 1/2$. A lift of norm at most 1 therefore exists, proving (i).

For (ii), the map

$$V : \ell_2^m(F) \longrightarrow L_1(\{-1, 1\}^m; F), \quad V(w_j) = \sum_j \delta_j w_j,$$

satisfies $\|Vw\|_{L_1} \geq (LD_f)^{-1} \|w\|_{\ell_2^m(F)}$. The functional

$$Vw \longmapsto b \sum_j y_j(w_j)$$

has norm at most $b\sqrt{m} LD_f = 1/2$ on its range. Hahn–Banach extends it to $L_1(F)$ with the same norm. Representing the extension in $L_\infty(F^*)$ gives Z and the stated moments. The amplification assertion follows from (2.6) and finite-sum duality. $\square$

3 A scalar Fourier-coefficient lemma

For a function on a sign cube, write

$$\mathcal{O}h(z) = \frac{h(z) - h(-z)}{2},$$

where $-z$ negates every coordinate. We call h odd if $\mathcal{O}h = h$. Fourier coefficients always use uniform measure. For an $m \times m$ sign matrix S and a binary matrix $P \in \{0, 1\}^{m \times m}$, put

$$S^P = \prod_{i,j} S_{ij}^{P_{ij}}, \quad \widehat{h}(P) = \mathbb{E}_S h(S) S^P.$$

For vector cubes we use binary indices $p \in \{0, 1\}^m$; $|p|$ denotes the number of ones and e_i the i -th unit vector. The notation σS multiplies row i by σ_i , and $S\tau$ multiplies column j by τ_j . We denote row i and column j by $S_{i,\cdot}$ and $S_{\cdot,j}$, respectively. All ranks in this section are over $\mathbb{F}_2$.

The point of the binary rank is the identity

$$\prod_{i,j} \chi(u_i, v_j)^{P_{ij}} = (-1)^{\sum_{i,j} P_{ij} u_i \cdot v_j}, \quad u_i, v_j \in \mathbb{F}_2^n.$$

An index of rank k yields k independent bilinear terms after binary changes of variables. In Section 4, this will turn a nonzero coefficient into a Walsh pairing in k times as many coordinates. We therefore need a coefficient whose index has rank at least two.

For example, take $S_{ij} = \chi(u_i, v_j)$ for $1 \leq i, j \leq 3$. The two indices with an odd number of ones

$$P_1 = \underbrace{\begin{pmatrix} 1 & 1 & 1 \\ 1 & 1 & 1 \\ 1 & 1 & 1 \end{pmatrix}}_{\text{binary rank 1}}, \quad S^{P_1} = \chi(u_1 + u_2 + u_3, v_1 + v_2 + v_3),$$

$$P_2 = \underbrace{\begin{pmatrix} 1 & 1 & 0 \\ 0 & 1 & 0 \\ 0 & 0 & 0 \end{pmatrix}}_{\text{binary rank 2}}, \quad S^{P_2} = \chi(u_1, v_1 + v_2) \chi(u_2, v_2)$$

yield one and two Walsh factors, respectively: $v_1 + v_2$ and v_2 vary independently.

Rank-one indices have rectangular support. The next lemma rules out an expansion supported only on odd rectangles when both families of prescribed moments are large enough.

Lemma 3.1 (A coefficient of rank at least two). *Let $M_0 = 4$, and suppose that*

$$4 \leq d = \lceil 2 \log_2 m \rceil < m.$$

Let $a, b > 0$, put $A = ma$, $B = mb$, and assume

$$\min(A, B) \geq 20M_0d(d+1)^2. \tag{3.1}$$

There is $c_ > 0$, depending only on m, a, b , with the following property. Suppose that*

$$\Psi : \{-1, 1\}^{m \times m} \rightarrow \mathbb{R}, \quad f, g : \{-1, 1\}^m \rightarrow \mathbb{R}$$

are odd, have sup norms at most M_0 , and satisfy

$$\widehat{f}(e_i) = a, \quad \widehat{g}(e_j) = b \tag{3.2}$$

together with the projection identities

$$\begin{aligned} \mathbb{E}_\sigma \sigma_i \Psi(\sigma S) &= a g(S_{i,:}), \\ \mathbb{E}_\tau \tau_j \Psi(S \tau) &= b f(S_{:,j}) \end{aligned} \tag{3.3}$$

for all i, j, S . Then

$$|\widehat{\Psi}(P)| \geq c_*$$

for some binary matrix P of rank at least two.

Proof. We first reduce the uniform lower bound to a nonvanishing assertion. If every index in the Fourier support of Ψ had rank at most one, a damped polynomial would encode their rectangular supports. Its prescribed mixed derivative will contradict a bound obtained by splitting it into two parts, each of low degree in one variable.

The admissible triples (Ψ, f, g) form a compact set in a finite-dimensional space. The maximum of $|\widehat{\Psi}(P)|$ over $\text{rank } P \geq 2$ is continuous. It therefore suffices to show that this maximum never vanishes. If the admissible set is empty, any positive c_* will do.

Suppose, to the contrary, that the Fourier support of Ψ has rank at most one. Oddness eliminates indices with an even number of ones. Every remaining index has a unique form $P = ph^t$ with $p, h \neq 0$, and both $k = |p|$ and $l = |h|$ are odd. Let

$$C_{kl} = \sum_{\substack{|p|=k \\ |h|=l}} \widehat{\Psi}(ph^t).$$

Parseval's identity and Cauchy-Schwarz give

$$|C_{kl}| \leq M_0 \sqrt{\binom{m}{k} \binom{m}{l}}. \quad (3.4)$$

To suppress rectangles that are large in both directions, consider the damped polynomial

$$H(s, u) = \sum_{\substack{1 \leq k, l \leq m \\ k, l \text{ odd}}} C_{kl} s^k u^l 2^{-kl}.$$

It has sup norm at most M_0 on $[-1, 1]^2$. To see this, take independent signs D_i, E_j, R_{ij} with means $s, u, 1/2$, respectively, and set $S_{ij} = D_i E_j R_{ij}$. For an odd rectangle ph^t ,

$$\mathbb{E} S^{ph^t} = s^{|p|} u^{|h|} 2^{-|p||h|},$$

so $H(s, u) = \mathbb{E} \Psi(S)$.

Delete from H the terms with both $k > d$ and $l > d$, obtaining H_0 . If $k, l > d$, then

$$(k + l) \log_2 m \leq kl, \quad kl/2 \geq k + l.$$

Using (3.4), the deleted part is bounded in sup norm by

$$M_0 \sum_{k, l > d} m^{(k+l)/2} 2^{-kl} \leq M_0 \sum_{k, l > d} 2^{-kl/2} \leq M_0 \sum_{k, l > d} 2^{-k-l} \leq M_0.$$

Consequently $\|H_0\|_\infty \leq 2M_0$.

For an odd rectangle ph^t , the vectors of row and column parities are p and h . Comparing Fourier coefficients in (3.3) therefore gives

$$C_{1l} = ma \sum_{|h|=l} \widehat{g}(h), \quad C_{k1} = mb \sum_{|p|=k} \widehat{f}(p).$$

The deletion defining H_0 does not affect terms with $k = 1$ or $l = 1$. Hence

$$\begin{aligned} \partial_s H_0(0, u) &= A \sum_h \widehat{g}(h) (u/2)^{|h|}, \\ \partial_u H_0(s, 0) &= B \sum_p \widehat{f}(p) (s/2)^{|p|}, \\ \partial_s \partial_u H_0(0, 0) &= AB/2. \end{aligned} \quad (3.5)$$

Independent signs of common mean $u/2$, or $s/2$, show that the first two expressions have sup norms at most AM_0 and BM_0 .

Every retained monomial has low degree in at least one variable. We separate these two possibilities with controlled sup norms. Expand H_0 in the Chebyshev basis $T_j(s)$, defined by $T_j(\cos \theta) = \cos(j\theta)$. Let H_1 retain the terms with $j \leq d$, and set $H_2 = H_0 - H_1$. Then

$$\deg_s H_1 \leq d, \quad \deg_u H_2 \leq d. \quad (3.6)$$

Indeed, every monomial of H_0 has s -degree at most d or u -degree at most d , and truncation in the Chebyshev basis annihilates every monomial of the former kind in H_2 .

Each Chebyshev coefficient of a polynomial on $[-1, 1]$ is bounded by twice its sup norm, by cosine orthogonality. Since $|T_j| \leq 1$,

$$\|H_1\|_\infty \leq 4M_0(d+1), \quad \|H_2\|_\infty \leq 6M_0(d+1).$$

Also, $|T'_j(0)| \leq j$, so every polynomial Q of degree at most d satisfies

$$|Q'(0)| \leq d(d+1)\|Q\|_\infty. \quad (3.7)$$

First apply (3.7) in s to H_1 , and in u to H_2 . Then (3.5) implies

$$\begin{aligned} \|\partial_s H_2(0, \cdot)\|_\infty &\leq AM_0 + 4M_0d(d+1)^2, \\ \|\partial_u H_1(\cdot, 0)\|_\infty &\leq BM_0 + 6M_0d(d+1)^2. \end{aligned}$$

By (3.6), these are polynomials of degree at most d in their displayed variables. Applying (3.7) once more and adding yields

$$|\partial_s \partial_u H_0(0, 0)| \leq M_0d(d+1)(A+B+10d(d+1)^2). \quad (3.8)$$

On the other hand, (3.1) gives

$$\frac{AB}{2} \geq 5M_0d(d+1)^2(A+B), \quad 10d(d+1)^2 \leq \frac{A+B}{4M_0}.$$

Since $5(d+1) > 1 + 1/(4M_0)$, these inequalities contradict (3.5) and (3.8). □

4 Uniform Walsh decay for finite-rank operators

4.1 Dimension growth

We now show that a slow decrease of Walsh norms forces a large Walsh norm at almost twice the original index. The proof uses the scalar coefficient lemma to find several independent Walsh factors inside a single pairing. To make that lemma applicable, we first arrange that our pairings depend only on their matrices of signs. The initial Walsh test is indexed by vectors, whereas the scalar lemma concerns arbitrary sign matrices; the frame construction below removes the extra dependence on the vectors.

An ordered linearly independent tuple in V_n will be called a *frame*. We identify covectors with vectors through the dot pairing. Independence is required separately on the two sides.

Lemma 4.1 (Pairs of frames). *The action*

$$(U, V) \longmapsto (HU, H^{-t}V), \quad H \in \text{GL}(n, \mathbb{F}_2),$$

is transitive on pairs of frames of fixed lengths with a prescribed matrix of pairings $(u_i \cdot v_j)$. Every binary $r \times s$ matrix is such a pairing matrix if $n \geq r + s$.

Proof. Consider two pairs (U, V) , (U', V') with the same pairing matrix. The covector frames define surjections $\pi, \pi' : V_n \rightarrow \mathbb{F}_2^s$. The isomorphism $u_i \mapsto u'_i$ between the primal spans respects evaluation, and hence maps $\text{span}(U) \cap \ker \pi$ onto $\text{span}(U') \cap \ker \pi'$. Extend this restriction to an isomorphism between the two kernels. Together with the map on the primal spans, it defines an isomorphism

$$\text{span}(U) + \ker \pi \longrightarrow \text{span}(U') + \ker \pi'$$

satisfying $\pi' H = \pi$. Complete it to an invertible map of V_n by choosing matching lifts of a basis complementary to the common evaluation image $\pi(\text{span}(U)) = \pi'(\text{span}(U'))$. Then $HU = U'$ and $\pi' H = \pi$, the latter being equivalent to $H^{-t}V = V'$. For feasibility in dimension $r + s$, given a matrix D , take $u_i = (e_i, 0)$ and $v_j = (D_{:,j}, e_j)$. $\square$

Lemma 4.2 (Growth from a slow decrease). *Fix an integer $m \geq 1$ and constants $D_g, D_f \geq 1$. Suppose that m and the parameters a, b in (2.9) satisfy the hypotheses of Lemma 3.1. There is $c_1 \in (0, 1/2]$, depending only on these parameters, such that the following holds. If*

$$c_m(G^*) \leq D_g, \quad c_m(F) \leq D_f, \quad 0 \neq T : G \rightarrow F,$$

and $n > 4m$ satisfies

$$w_{n-2m}(T) \leq 2w_n(T), \tag{4.1}$$

then

$$w_{2(n-2m)}(T) \geq c_1 w_n(T). \tag{4.2}$$

The operator T need not have finite rank.

Proof. Put $n' = n - 2m$. We will turn a Walsh test at index n into bounded scalar functions satisfying Lemma 3.1. A coefficient of binary rank $k \geq 2$ will then give a Walsh test at index kn' .

Step 1: realize the Walsh signs exactly and equivariantly. Choose families $p_u \in G$, $q_v \in F^*$ admissible in (2.1), changing a sign if necessary, such that

$$\gamma = \mathbb{E}_{u,v} \chi(u, v) q_v(T p_u) \geq w_n(T)/2 > 0.$$

Amplify T over $V_n \times V_n$, and define

$$x'_u(\alpha, \beta) = (-1)^{u \cdot \beta + \alpha \cdot \beta} p_{u+\alpha}, \quad y'_v(\alpha, \beta) = (-1)^{\alpha \cdot v} q_{v+\beta}.$$

Uniform translation shows that every x'_u and y'_v has norm at most one in the corresponding finite L_2 space. Writing T' for the amplified operator, we obtain

$$y'_v(T' x'_u) = \gamma \chi(u, v).$$

Indeed, the pairing includes expectation over α, β , and its phase is

$$u \cdot \beta + \alpha \cdot \beta + \alpha \cdot v = (u + \alpha) \cdot (v + \beta) + u \cdot v.$$

Amplify once more over $\mathcal{G} = \text{GL}(n, \mathbb{F}_2)$, obtaining $\tilde{T} : \tilde{G} \rightarrow \tilde{F}$, and set

$$x_u(R) = x'_{Ru}, \quad y_v(R) = y'_{R^{-t}v}.$$

For the bilinear form $B(x, y) = y(\tilde{T}x)$, this gives

$$x_u \in B_{\tilde{G}}, \quad y_v \in B_{\tilde{F}^*}, \quad B(x_u, y_v) = \gamma \chi(u, v). \tag{4.3}$$

Use the same right translations $\Lambda(H)z(R) = z(RH)$ on the domain and target dual. They are isometries, preserve B under simultaneous translation, and satisfy

$$\Lambda(H)x_u = x_{Hu}, \quad \Lambda(H)y_v = y_{H^{-t}v}, \quad \Lambda(H_1)\Lambda(H_2) = \Lambda(H_1H_2).$$

The orders follow directly from $(RH)^{-t} = R^{-t}H^{-t}$. By Lemma 2.1, $w_j(\tilde{T}) = w_j(T)$ for every j . Equation (2.6) also preserves the bounds D_g, D_f , so the moment lifts remain available with the same a, b .

Step 2: lift frames while preserving the symmetry. For every primal m -frame $U = (u_i)_{i=1}^m$, choose $P_{U,\varepsilon} \in B_{\tilde{G}}$ with

$$\mathbb{E}_\varepsilon \varepsilon_i P_{U,\varepsilon} = ax_{u_i}.$$

We can make these choices equivariant. For one reference frame, take the lifts from Lemma 2.3 and average each of them over its stabilizer under Λ . The prescribed moments are fixed by this stabilizer, and the averaged lifts are stabilizer-invariant. Transport them to any other frame by $\Lambda(H)$. Two transporters differ on the right by a stabilizer element, so the result is well-defined and equivariant.

The same construction for covector frames, using the action H^{-t} , gives $Z_{V,\delta} \in B_{\tilde{F}^*}$ with

$$\mathbb{E}_\delta \delta_j Z_{V,\delta} = by_{v_j}.$$

The two systems of lifts can be chosen independently. Their equivariance, invariance of B , and Lemma 4.1 show that the following pairings depend only on the displayed sign patterns and lift labels:

$$\begin{aligned} \Phi(S, \varepsilon, \delta) &= B(P_{U,\varepsilon}, Z_{V,\delta}), & S_{ij} &= \chi(u_i, v_j), \\ \phi(s, \varepsilon) &= B(P_{U,\varepsilon}, y_v), & s_i &= \chi(u_i, v), \quad v \neq 0, \\ \psi(t, \delta) &= B(x_u, Z_{V,\delta}), & t_j &= \chi(u, v_j), \quad u \neq 0. \end{aligned} \tag{4.4}$$

All patterns are feasible since $n > 4m$. The moment identities and (4.3) give

$$\begin{aligned} \mathbb{E}_\varepsilon \varepsilon_i \Phi(S, \varepsilon, \delta) &= a\psi(S_{i,:}, \delta), & \mathbb{E}_\delta \delta_j \Phi(S, \varepsilon, \delta) &= b\phi(S_{:,j}, \varepsilon), \\ \mathbb{E}_\varepsilon \varepsilon_i \phi(s, \varepsilon) &= a\gamma s_i, & \mathbb{E}_\delta \delta_j \psi(t, \delta) &= b\gamma t_j. \end{aligned} \tag{4.5}$$

Step 3: control the odd parts on a smaller Walsh cube. We need a bound that remains useful after division by γ . For every fixed choice of lift labels, we claim that

$$\|\mathcal{O}_S \Phi(\cdot, \varepsilon, \delta)\|_\infty, \quad \|\mathcal{O}_s \phi(\cdot, \varepsilon)\|_\infty, \quad \|\mathcal{O}_t \psi(\cdot, \delta)\|_\infty \leq w_{n'}(T). \tag{4.6}$$

Decompose $V_n = V_{n'} \oplus \mathbb{F}_2^m \oplus \mathbb{F}_2^m$, and fix a sign matrix $S_0 = ((-1)^{D_{ij}})$. For $u, v \in V_{n'}$, take the primal frame

$$U(u)_i = (u, e_i, 0)$$

and the two covector frames

$$V^+(v)_j = (v, D_{:,j}, e_j), \quad V^-(v)_j = (v, D_{:,j} + \mathbf{1}, e_j),$$

where $\mathbf{1} \in \mathbb{F}_2^m$ is the all-ones vector. The tags e_i, e_j ensure independence for every u, v . Their sign matrices are $\chi(u, v)S_0$ and $-\chi(u, v)S_0$, so

$$B\left(P_{U(u),\varepsilon}, \frac{Z_{V^+(v),\delta} - Z_{V^-(v),\delta}}{2}\right) = \chi(u, v)(\mathcal{O}_S \Phi)(S_0, \varepsilon, \delta).$$

Both vector families lie in their respective unit balls, and each depends only on its own Walsh index. Testing (2.1) at n' , and using amplification invariance, proves the first bound.

For ϕ , retain $U(u)$ and use the nonzero covectors (v, h, e_1) , $(v, h + \mathbf{1}, e_1)$, where $s_i = (-1)^{h_i}$. Half the difference of the corresponding y 's gives the same calculation. For ψ , use the covector frame $(v, 0, e_j)$ and half the difference of the primal vectors at (u, e_1, h) , $(u, e_1, h + \mathbf{1})$. Their pairings are $\chi(u, v)(-1)^{h_j}$ and its negative. The e_1 tags keep the singleton vectors nonzero. This proves (4.6).

Step 4: find a Fourier coefficient of binary rank at least two. Define

$$\begin{aligned} f(s) &= \gamma^{-1} \mathcal{O}_s \mathbb{E}_\varepsilon \phi(\varepsilon s, \varepsilon), \\ g(t) &= \gamma^{-1} \mathcal{O}_t \mathbb{E}_\delta \psi(\delta t, \delta), \\ \Psi(S) &= \gamma^{-1} \mathcal{O}_S \mathbb{E}_{\varepsilon, \delta} \Phi(\varepsilon S \delta, \varepsilon, \delta). \end{aligned} \tag{4.7}$$

These functions are odd, and the slow-decrease hypothesis gives

$$\max(\|f\|_\infty, \|g\|_\infty, \|\Psi\|_\infty) \leq w_{n-2m}(T)/\gamma \leq 4.$$

Changing variables $s' = \varepsilon s$ in the first coefficient and using (4.5), we obtain

$$\widehat{f}(e_i) = \gamma^{-1} \mathbb{E}_{s', \varepsilon} s'_i \varepsilon_i \phi(s', \varepsilon) = a.$$

The odd projection leaves this coefficient unchanged. Likewise, $\widehat{g}(e_j) = b$.

We next check the row and column projections required by Lemma 3.1. A first row projection is unchanged by applying global odd projection, since

$$\mathbb{E}_\sigma \sigma_i h(-\sigma S) = -\mathbb{E}_\sigma \sigma_i h(\sigma S).$$

Substitute $\xi = \varepsilon \sigma$; then ξ, ε, δ are still independent uniform sign vectors. By (4.5),

$$\begin{aligned} \gamma \mathbb{E}_\sigma \sigma_i \Psi(\sigma S) &= \mathbb{E}_{\xi, \varepsilon, \delta} \xi_i \varepsilon_i \Phi(\xi S \delta, \varepsilon, \delta) \\ &= a \mathbb{E}_{\xi, \delta} \xi_i \psi(\xi_i(\delta S_{i,:}), \delta) \\ &= a \mathbb{E}_\delta \frac{\psi(\delta S_{i,:}, \delta) - \psi(-\delta S_{i,:}, \delta)}{2} \\ &= a \gamma g(S_{i,:}). \end{aligned}$$

Interchanging rows and columns, and simultaneously (a, ε, ϕ) with (b, δ, ψ) , gives $\mathbb{E}_\tau \tau_j \Psi(S \tau) = b f(S_{:,j})$. Thus all hypotheses of Lemma 3.1 hold, with the fixed parameters m, a, b .

Choose a binary matrix P of rank $k \geq 2$ with $|\widehat{\Psi}(P)| \geq c_*$. Since Ψ is odd and this coefficient is nonzero, the total number of ones in P is odd. Its coefficient is therefore unchanged when the odd projection is removed from (4.7). Writing $\Phi_{\varepsilon, \delta}(S) = \Phi(S, \varepsilon, \delta)$, row and column sign multiplication gives the exact identity

$$\gamma \widehat{\Psi}(P) = \mathbb{E}_{\varepsilon, \delta} \left[\left(\prod_{i,j} (\varepsilon_i \delta_j)^{P_{ij}} \right) \widehat{\Phi}_{\varepsilon, \delta}(P) \right].$$

For some fixed ε, δ , therefore, $h = \Phi_{\varepsilon, \delta}$ satisfies

$$|\widehat{h}(P)| \geq c_* \gamma. \tag{4.8}$$

No oddness assertion about this individual function h is needed.

Step 5: extract the independent Walsh factors. Keep P, ε, δ fixed. For arbitrary tuples $(u'_i)_{i=1}^m, (v'_j)_{j=1}^m$ in $V_{n'}$, set

$$U_i = (u'_i, e_i, 0), \quad V(D)_j = (v'_j, D_{:,j}, e_j),$$

where D ranges over binary $m \times m$ matrices. The tagged tuples are frames even if the original tuples have repeated or zero entries. The Fourier coefficient formula gives

$$B\left(P_{U,\varepsilon}, \mathbb{E}_D(-1)^{\langle P,D \rangle} Z_{V(D),\delta}\right) = \widehat{h}(P)(-1)^{\sum_{i,j} P_{ij} u'_i \cdot v'_j}, \quad (4.9)$$

where $\langle P, D \rangle = \sum_{i,j} P_{ij} D_{ij}$ is computed modulo two. Indeed, translate the uniform matrix D by $(u'_i \cdot v'_j)_{ij}$ in the defining Fourier average. Both vectors in (4.9) have norm at most one. The first depends only on the u' -tuple, and the second only on the v' -tuple.

Since $\text{rank}_{\mathbb{F}_2} P = k$, choose binary $m \times k$ matrices I, J with $I^t P J = \text{Id}_k$. For $\zeta, \eta \in (V_{n'})^k$, substitute

$$u'_i = \sum_{\ell=1}^k I_{i\ell} \zeta_\ell, \quad v'_j = \sum_{\ell=1}^k J_{j\ell} \eta_\ell.$$

The exponent in (4.9) becomes $\sum_{\ell=1}^k \zeta_\ell \cdot \eta_\ell$. We have thus obtained a Walsh test on $V_{kn'}$, of amplitude $\widehat{h}(P)$, with separate unit-ball families on its two sides. Monotonicity and amplification invariance now yield

$$w_{2n'}(T) \geq w_{kn'}(T) = w_{kn'}(\widetilde{T}) \geq |\widehat{h}(P)| \geq (c_*/2)w_n(T).$$

Taking $c_1 = \min(1/2, c_*/2)$ proves the lemma. $\square$

4.2 Uniform decay

The growth lemma turns a sufficiently late slow decrease into an infinite sequence of slow decreases for the same operator. For a finite-rank operator, this would contradict its Hilbert-space bound. This is the only point at which finite rank is used.

Theorem 4.3 (Uniform finite-rank decay). *Fix m, D_g, D_f satisfying the assumptions of Lemma 4.2. There is an integer n_0 , depending only on these parameters, such that every finite-rank $T : G \rightarrow F$ with $c_m(G^*) \leq D_g$ and $c_m(F) \leq D_f$ satisfies*

$$w_n(T) \leq \frac{1}{2} w_{n-2m}(T) \quad (n \geq n_0). \quad (4.10)$$

In particular there are numbers $\rho_N \rightarrow 0$, independent of G, F, T , such that

$$w_N(T) \leq \rho_N \|T\|. \quad (4.11)$$

Both bounds also hold for every $T : G \rightarrow F$ that is an operator-norm limit of finite-rank maps, under the same moment hypotheses on G^ and F .*

Proof. Let c_1 be given by Lemma 4.2. Choose an integer $K > \log_2(1/c_1)$, and put

$$n_0 = 4m(K + 2) + 2.$$

Fix a nonzero finite-rank T , and suppose that (4.1) holds at some $n \geq n_0$. The growth lemma gives

$$w_{2n-4m}(T) \geq c_1 w_n(T).$$

By monotonicity, the same lower bound holds throughout $[n, 2n - 4m]$.

Set $a_n = \lceil 3n/2 \rceil$. The K consecutive blocks

$$[a_n + 2m(j-1), a_n + 2mj] \quad (1 \leq j \leq K)$$

fit in this interval, since $2n - a_n = \lfloor n/2 \rfloor \geq 2mK + 4m + 1$. Their endpoint ratios telescope:

$$\prod_{j=1}^K \frac{w_{a_n+2m(j-1)}(T)}{w_{a_n+2mj}(T)} = \frac{w_{a_n}(T)}{w_{a_n+2mK}(T)} \leq \frac{1}{c_1}.$$

Since $2^K > 1/c_1$, one right endpoint t satisfies

$$w_{t-2m}(T) \leq 2w_t(T), \quad t \geq 3n/2, \quad w_t(T) \geq c_1 w_n(T).$$

We can repeat the argument at t . Every application of the growth lemma returns an inequality for the original T ; its auxiliary amplifications are discarded. We therefore obtain indices $n_j \geq (3/2)^j n$ with $w_{n_j}(T) \geq c_1^j w_n(T) > 0$. The rank-dependent estimate (2.2), for this one fixed operator, implies

$$0 < w_n(T) \leq D_T c_1^{-j} 2^{-n_j/2} \longrightarrow 0,$$

a contradiction. Thus (4.1) is impossible for $n \geq n_0$, proving the strict form of (4.10) for nonzero finite-rank T . The zero operator satisfies the asserted weak inequality.

Iteration proves (4.11); for example, take

$$\rho_N = \begin{cases} 1, & N < n_0, \\ 2^{-\lfloor (N-n_0)/(2m) \rfloor}, & N \geq n_0. \end{cases}$$

Finally, if finite-rank maps T_j converge in operator norm to T , then (2.3) gives $w_s(T_j) \rightarrow w_s(T)$ for every s , while $\|T_j\| \rightarrow \|T\|$. Passing to the limit proves both estimates for T . $\square$

Remark 4.4. The constant D_T may depend on the operator rank. It is used only to rule out an infinite sequence of slowly decreasing indices for one fixed operator; the resulting n_0 and ρ_N depend only on the fixed moment parameters. The binary ranks of Fourier indices in Lemmas 3.1 and 4.2 play a different role: they count the independent Walsh factors extracted from a coefficient.

5 Compact factorization and the approximation property

We now transfer Walsh decay to the identity of X . The approximation property will let us apply Theorem 4.3 to compact maps into X . We therefore factor the inclusion of each finite-dimensional $E \subset X$ through a compact map whose domain dual has a controlled moment estimate. Using E itself as the domain would not suffice: the quotient E^* of X^* need not inherit the cotype estimate of X^* . Instead, we enlarge E to retain the moment lifts available in X , and use decreasing weights to obtain compactness.

Lemma 5.1 (A compact domain with controlled dual moments). *Let $m \geq 1$ be an integer, let $\Delta \geq 1$, and let X be a Banach space with $c_m(X^*) \leq \Delta$. For every nonzero finite-dimensional subspace $E \subset X$, there exist a Banach space G , an injective compact operator $J : G \rightarrow X$, and a contraction $\iota : E \rightarrow G$, such that*

$$J\iota x = x \quad (x \in E), \quad \|J\| \leq 2, \quad c_m(G^*) \leq 12\Delta. \quad (5.1)$$

Proof. Finite-dimensional enlargements. We construct nested finite-dimensional subspaces

$$E = E_0 \subset E_1 \subset E_2 \subset \cdots \subset X$$

such that each $z = (z_i)_{i=1}^m \in \ell_2^m(E_l)$ has a lift $H_l : \{-1, 1\}^m \rightarrow E_{l+1}$ satisfying

$$\mathbb{E}_\varepsilon \varepsilon_i H_l(\varepsilon) = z_i, \quad \|H_l\|_{L_2(E_{l+1})} \leq 3\Delta \|z\|_{\ell_2^m(E_l)}. \quad (5.2)$$

First take moments in the whole space X . The map

$$U : L_2(\{-1, 1\}^m; X) \rightarrow \ell_2^m(X), \quad UH = (\mathbb{E}_\varepsilon \varepsilon_i H(\varepsilon))_i,$$

has the bounded right inverse $(z_i) \mapsto \sum_i \varepsilon_i z_i$, and

$$\|U^*(\theta_i)\|_{L_2(X^*)} = \left(\mathbb{E}_\varepsilon \left\| \sum_i \varepsilon_i \theta_i \right\|^2 \right)^{1/2} \geq \Delta^{-1} \|(\theta_i)\|_{\ell_2^m(X^*)}.$$

By (2.8), the infimal lift norm of a unit tuple is at most Δ , so a lift of norm below 2Δ exists. Take a finite η -net on the unit sphere of $\ell_2^m(E_l)$, where $0 < \eta \leq \min(1/2, \Delta/\sqrt{m})$, and choose such a lift for each net point. Let E_{l+1} be the span of E_l and all the values of these finitely many lifts.

For any unit tuple $z \in \ell_2^m(E_l)$, choose a net point v with $\|z - v\|_{\ell_2^m(X)} \leq \eta$. Add $\sum_i \varepsilon_i(z_i - v_i)$ to its lift. This corrects the moments exactly and adds at most

$$\left\| \sum_i \varepsilon_i(z_i - v_i) \right\|_{L_2(X)} \leq \sqrt{m} \|z - v\|_{\ell_2^m(X)} \leq \Delta$$

to the norm. This proves (5.2) for unit tuples; scaling and the zero lift cover the other cases.

The compact map. Put

$$\mathcal{H} = \left(\bigoplus_{l=0}^{\infty} E_l \right)_{\ell_2}, \quad j : \mathcal{H} \rightarrow X, \quad j(z) = \sum_{l \geq 0} 2^{-l} z_l.$$

Cauchy–Schwarz proves absolute convergence of the series and gives

$$\|j\| \leq \left(\sum_{l \geq 0} 4^{-l} \right)^{1/2} < 2.$$

The same estimate on the tails shows that the finite-rank truncations converge in operator norm. Thus j is compact. Let $G = \mathcal{H}/\ker j$, with quotient map Q , and define J by $JQ = j$. The kernel is closed, so G is Banach; the quotient norm gives $\|J\| \leq \|j\| < 2$, and J is injective. Every element of B_G has a representative in $2B_{\mathcal{H}}$. Hence $J(B_G) \subset j(2B_{\mathcal{H}})$ is relatively compact, which proves compactness of J . Finally, $\iota(x) = Q(x, 0, 0, \dots)$ is a contraction and $J\iota x = x$. The finite-rank truncations of j need not annihilate $\ker j$, so they need not induce operators on G . Thus this compactness argument does not itself approximate J by finite-rank maps; that passage will use the approximation property of X .

Dual moments from the weighted shift. The nesting of the E_l 's makes

$$R : \mathcal{H} \rightarrow \mathcal{H}, \quad R(z_0, z_1, \dots) = (0, 2z_0, 2z_1, \dots)$$

a bounded map. The factor 2 compensates for the next dyadic weight, so $jR = j$ and therefore $QR = Q$. This identity lets us assemble the finite-level lifts into exact moment lifts in G .

For $g_1, \dots, g_m \in G$, choose representatives $z^i = (z_l^i)_l \in \mathcal{H}$ with $Qz^i = g_i$ and $\|z^i\| \leq 2\|g_i\|$, taking $z^i = 0$ when $g_i = 0$. Use (5.2) on each tuple $(z_l^i)_i$, and set

$$\tilde{H}(\varepsilon) = (0, 2H_0(\varepsilon), 2H_1(\varepsilon), \dots).$$

Summing squared estimates yields

$$\|\tilde{H}\|_{L_2(\mathcal{H})} \leq 6\Delta \left(\sum_i \|z^i\|^2 \right)^{1/2} \leq 12\Delta \left(\sum_i \|g_i\|^2 \right)^{1/2}.$$

Every atom of the sign cube has positive mass, so this estimate also shows that $\tilde{H}(\varepsilon) \in \mathcal{H}$ for every ε . Its i -th moment is Rz^i . Since $QR = Q$,

$$\mathbb{E}_\varepsilon \varepsilon_i Q\tilde{H}(\varepsilon) = QRz^i = Qz^i = g_i.$$

For $\theta_1, \dots, \theta_m \in G^*$, it follows that

$$\begin{aligned} \left| \sum_i \theta_i(g_i) \right| &= \left| \mathbb{E}_\varepsilon \left\langle Q\tilde{H}(\varepsilon), \sum_i \varepsilon_i \theta_i \right\rangle \right| \\ &\leq 12\Delta \left(\sum_i \|g_i\|^2 \right)^{1/2} \left(\mathbb{E}_\varepsilon \left\| \sum_i \varepsilon_i \theta_i \right\|^2 \right)^{1/2}. \end{aligned}$$

Taking the supremum over $\sum_i \|g_i\|^2 \leq 1$ gives $c_m(G^*) \leq 12\Delta$ by finite ℓ_2 duality. $\square$

We now complete the proof of the main theorem.

Proof of Theorem 1.1. First suppose $K(X) < \infty$. On each finite cube, the adjoint of the Rademacher projection on $L_2(X)$ is the corresponding projection on $L_2(X^*)$. Therefore $K(X^*) = K(X)$. By the classical characterization recalled in the introduction, both X and X^* have type strictly greater than one.

If a space Y has type $p > 1$, with constant D , then Y^* has cotype $p' = p/(p-1)$. Indeed, for finite tuples $y_i \in Y$, $y_i^* \in Y^*$,

$$\begin{aligned} \left| \sum_i y_i^*(y_i) \right| &= \left| \mathbb{E}_\varepsilon \left\langle \sum_i \varepsilon_i y_i, \sum_i \varepsilon_i y_i^* \right\rangle \right| \\ &\leq D \left(\sum_i \|y_i\|^p \right)^{1/p} \left(\mathbb{E}_\varepsilon \left\| \sum_i \varepsilon_i y_i^* \right\|^2 \right)^{1/2}. \end{aligned}$$

Taking the supremum over the unit ball of $\ell_p^m(Y)$ proves the claim. Apply it to $Y = X$ and $Y = X^*$, and restrict the resulting cotype inequality on X^{**} to X . This direction does not use the approximation property.

Conversely, suppose that X has the approximation property and

$$C_q(X) \leq C_F, \quad C_r(X^*) \leq C_Y$$

for finite $q, r \geq 2$ and finite constants $C_F, C_Y \geq 1$. For an integer m , set

$$D_f = C_F m^{1/2-1/q}, \quad \Delta = C_Y m^{1/2-1/r}, \quad D_g = 12\Delta.$$

The parameters in (2.9) then satisfy

$$ma = \frac{m^{1/r}}{24LC_Y}, \quad mb = \frac{m^{1/q}}{2LC_F}.$$

These positive powers of m eventually exceed any fixed multiple of $(\log m)^3$. Choose m once so that the hypotheses of Lemma 3.1 hold, and fix the decay sequence ρ_N given by Theorem 4.3 for m, D_g, D_f . All these parameters are independent of the finite-dimensional subspaces and approximating operators chosen below.

We first show that

$$w_N(\text{Id}_X) \leq 2\rho_N \quad (N \geq 0). \quad (5.3)$$

For a nonzero finite-dimensional $E \subset X$, choose $J : G \rightarrow X$ and $\iota : E \rightarrow G$ by Lemma 5.1. Apply the approximation property to the compact set $M = \overline{J(B_G)}$. For every $j \geq 1$, it supplies a bounded finite-rank map $S_j : X \rightarrow X$ such that

$$\|S_j J - J\| \leq \sup_{x \in M} \|S_j x - x\| < 1/j.$$

Thus J is an operator-norm limit of finite-rank maps from G to X . Since $c_m(G^*) \leq D_g$ and $c_m(X) \leq D_f$, the norm-limit assertion of Theorem 4.3 gives

$$w_N(J) \leq \rho_N \|J\| \leq 2\rho_N. \quad (5.4)$$

Only the composites $S_j J$ converge in operator norm; no common bound on $\|S_j\|$ is required.

Now fix any input $x = (x_u)_{u \in V_N} \in L_2(V_N; X)$. If $x = 0$, there is nothing to prove. Otherwise take $E = \text{span}\{x_u : u \in V_N\}$ and choose the corresponding J, ι . The family $z_u = \iota(x_u)$ satisfies $\|z\|_{L_2(G)} \leq \|x\|_{L_2(X)}$ and $Jz_u = x_u$. Consequently,

$$\|W_N^{\text{Id}_X} x\|_{L_2(X)} = \|W_N^J z\|_{L_2(X)} \leq 2\rho_N \|x\|_{L_2(X)},$$

which proves (5.3).

If $K(X) = \infty$, the characterization (1.1) supplies $\lambda \geq 1$ such that, for every $N \geq 1$, there are $x_u \in B_X$, $u \in V_N$, with

$$\left\| \sum_{u \in V_N} \alpha_u x_u \right\| \geq \lambda^{-1} \sum_{u \in V_N} |\alpha_u|$$

for all real coefficients. These vectors form an input of L_2 norm at most one, while every Walsh output satisfies

$$\|(W_N^{\text{Id}_X} x)(v)\| = \left\| 2^{-N} \sum_{u \in V_N} \chi(u, v) x_u \right\| \geq \lambda^{-1}.$$

Thus $w_N(\text{Id}_X) \geq \lambda^{-1}$ for every N , contradicting (5.3) and $\rho_N \rightarrow 0$. This proves $K(X) < \infty$. $\square$

Proof of Corollary 1.2. If no uniform κ existed, choose spaces X_j with the stated fixed cotype bounds and AP, but $K(X_j) \rightarrow \infty$, and set

$$Z = \left(\bigoplus_{j \geq 1} X_j \right)_{\ell_2}.$$

Finite-coordinate truncations converge uniformly on each compact subset of Z . Approximating its finitely many compact coordinate images by finite-rank maps therefore proves AP for Z ; no uniform bound on the norms of those maps is needed. Moreover, $Z^* = (\oplus_{j \geq 1} X_j^*)_{\ell_2}$ isometrically. For any finite tuple $z_i = (z_{ij})_j$ in Z , Minkowski's inequality in $\ell_{q/2}$ and the coordinate cotype bounds give

$$\left(\sum_i \|z_i\|_Z^q \right)^{2/q} \leq \sum_j \left(\sum_i \|z_{ij}\|^q \right)^{2/q} \leq C^2 \mathbb{E}_\varepsilon \left\| \sum_i \varepsilon_i z_i \right\|_Z^2.$$

Thus $C_q(Z) \leq C$, and the same argument on the dual gives $C_r(Z^*) \leq D$. Theorem 1.1 implies $K(Z) < \infty$. Restriction of each finite-cube Rademacher projection to a coordinate summand gives $K(X_j) \leq K(Z)$, a contradiction.

For the entropy assertion, finite-dimensional gauge spaces have AP, so the first assertion bounds their K -constants by κ . The bounded- K entropy-duality theorem of Artstein–Milman–Szarek–Tomczak-Jaegermann [1, Theorem 5], in the covering-number formulation of [8, Theorem 1.1, equation (1.6), and Section 2], has constants $a, b \geq 1$ depending only on an upper bound for the K -convexity constant of either gauge space. Polarity identifies $X_{B^\circ} = X_B^*$, and the finite-cube adjoint identity used above gives $K(X_B^*) = K(X_B)$. Thus the same bound is available for the polar bodies in the reverse inequality. Also, X_{tL} is linearly isometric to X_L . Apply the covering theorem to K, tL . Since $(tL)^\circ = t^{-1}L^\circ$ and $N(t^{-1}L^\circ, cK^\circ) = N(L^\circ, ctK^\circ)$ for $c > 0$, both inequalities in (1.2) follow for every $t > 0$, with constants depending only on κ . $\square$

6 The conjecture on Grothendieck pairs

We now apply the characterization to the Grothendieck-pair conjecture. The deduction through complemented Euclidean subspaces is the one in [6, Corollary 4.8]; the main theorem allows ordinary AP here. For an algebraic tensor $t \in E \otimes F$, recall the injective and projective norms

$$\|t\|_\varepsilon = \sup_{\substack{x^* \in B_{E^*} \\ y^* \in B_{F^*}}} |(x^* \otimes y^*)(t)|, \quad \|t\|_\pi = \inf_{t = \sum_i x_i \otimes y_i} \sum_i \|x_i\| \|y_i\|.$$

Their completions are denoted by $E \widehat{\otimes}_\varepsilon F$ and $E \widehat{\otimes}_\pi F$. Both norms satisfy the metric mapping property: tensoring bounded maps multiplies the norm by at most the product of their operator norms. This follows directly from the two definitions.

Let $\mathcal{E} = (E_n)_{n \geq 1}$ be a sequence of real Banach spaces with $\dim E_n = n$. Following [6, Definition 1.2], call $(\mathcal{E}, F)$ a *Grothendieck pair* if there is $C < \infty$ such that

$$\|A \otimes \text{Id}_F : E_m \widehat{\otimes}_\varepsilon F \longrightarrow E_n^* \widehat{\otimes}_\pi F\| \leq C \|A\| \quad (6.1)$$

for all $m, n \geq 1$ and every $A : E_m \rightarrow E_n^*$. The existence of the bounded extension in (6.1) is part of the condition.

Corollary 6.1. *If $(\mathcal{E}, F)$ is a Grothendieck pair and F is infinite-dimensional with the approximation property, then $(\mathcal{E}, \ell_2)$ is a Grothendieck pair. In particular, Pisier's conjecture [6, Conjecture 1.4] holds, with the approximation property in place of the bounded approximation property.*

Proof. By Gupta–Misra–Ray [6, Theorem 1.3], the pair condition implies finite cotype of F and F^* . Thus Theorem 1.1 makes F K -convex.

We use the complemented Euclidean subspace theorem of Figiel–Tomczak-Jaegermann [5], in the formulation [14, Theorem 4]: for an infinite-dimensional K -convex space F , there is $D < \infty$ such that for every $d \geq 1$ there are bounded maps

$$U_d : \ell_2^d \rightarrow F, \quad V_d : F \rightarrow \ell_2^d, \quad V_d U_d = \text{Id}_{\ell_2^d}, \quad \|U_d\| \|V_d\| \leq D.$$

By the metric mapping property and the factorization

$$A \otimes \text{Id}_{\ell_2^d} = (\text{Id}_{E_n^*} \otimes V_d)(A \otimes \text{Id}_F)(\text{Id}_{E_m} \otimes U_d),$$

the corresponding tensor operator with ℓ_2^d in place of F has norm at most $CD\|A\|$, uniformly in d, m, n, A .

Every algebraic tensor in $E_m \otimes \ell_2$ has its second coordinates in a finite-dimensional Hilbert subspace $H \subset \ell_2$. The inclusion of H and the orthogonal projection onto H both have norm one. The metric mapping property therefore shows that both tensor norms on tensors supported in H agree with the corresponding norms computed in ℓ_2 . The uniform estimate for finite-dimensional Hilbert spaces consequently holds on $E_m \otimes \ell_2$, and extends to its injective completion. This is (6.1) for ℓ_2 , with constant CD . $\square$

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